

Modeling the cutting edge geometry of scalpel blades

Pralav P Shetty¹, Ryan W Hatton¹, Andrew C Barnett¹, Andrew J Homich¹, Jason Z Moore¹

¹Department of Mechanical and Nuclear Engineering, The Pennsylvania State University,
USA

Corresponding author:

Jason Z Moore, 318 Leonhard Building, University Park, PA 16802, USA.

Email: jzm14@psu.edu

Abstract

Scalpel blades are commonly used in surgery to perform invasive medical procedures, yet there has been limited research on the geometry that makes up these cutting instruments. Improved understanding of a scalpel's cutting edge geometry can lead to smaller and more effective tissue cutting instruments. The following study develops a generalized geometric model that describes the cutting edge geometry in terms of rake and inclination angle of any continuously differentiable scalpel cutting edge surface. The parameter of scalpel-tissue contact area is also examined. The geometry of commonly used scalpel blades (#10, #11, #12 and #15) are compared to each other and their cutting force through phantom gel measured. It was found that blade #10 displayed the lowest

total cutting force followed by blade #15, #11 and #12 respectively. Blade #10 also displayed the lowest normalized cutting force followed by blade #15, #12 and #11 respectively.

Keywords

Cutting, Surfaces

Introduction

Scalpel blades are commonly used handheld instruments designed to cut tissue effectively to perform various invasive surgical procedures. These procedures include those that end with the suffix “ectomy” which means “to cut.” Common examples of this would be lumpectomy (removal of cancerous breast tissue) and prostatectomy (removal of the prostate). These are cutting processes where soft tissue is the work material and the scalpel blade acts as the tool. The goals of all these procedures are to cut tissue accurately and to minimize trauma to the patient. In order to reach these goals, sharp edged cutting tools apply cutting forces to the tissue to fracture the bonds that hold cells in the tissue together.

Scalpels used in medicine are designated with a blade number which allows for the identification of the cutting edge shape. Figure 1 displays common industry scalpels which

are used in this study. Varying the scalpel blade cutting direction allows for the use of three basic incision methods: press cutting, slide cutting and scrape cutting, as shown in Figure 2. Utilizing these methods, tissue can be accurately removed for a wide variety of invasive surgical procedures.

Accurate cutting of soft tissue presents a challenge in scalpel procedures because the large cutting forces cause high tissue deflection, which degrades the accuracy of the cut depth. Poor cutting accuracy can lead to lower procedural efficacy and reduced quality of life for the patient. For example, in a lumpectomy procedure where cancerous breast tissue is cut out of the body, the surgeon aims to remove the cancerous tissue with a rim of healthy tissue around it, called the margins. After the procedure the margins are tested for cancer. In 20% to 40% of patients positive margins are found (margins contain cancerous tissue) requiring the need for additional mutilating surgical procedures to obtain tumor free margins ¹. These additional surgeries lead to a negative effect on cosmesis, increased wound infection risk and a decreased patient confidence in the health care treatment being received ². As another example, in a prostatectomy, inaccuracies in cutting can lead to increased side effects including impotence and incontinence ^{3, 4}. Improved scalpel cutting accuracy can have a great benefit to these and other cutting procedures.

Large cutting forces generated during tissue cutting limit the design of instruments used in Minimally Invasive Surgery (MIS). There is a continual drive in MIS procedures to reduce the number and size of the abdominal incisions by using smaller instruments to reduce pain, shorten length of stay in hospital, allow for faster return to activity and create better cosmetic outcomes ⁵. However, these small instruments must be made large enough to withstand the high tissue cutting forces they experience in operation. More effective cutting edge design can reduce the tissue cutting force and allow for the development of smaller MIS instruments.

Much of the work currently performed on tissue cutting instruments have examined the topics of intraoperative blood loss, operative time, operative cost and post-operative healing time of various tissue cutting instruments including scalpels, harmonic scalpels and electrocautery devices ⁶⁻⁸. *Chanthasopeephan et al.* ⁹ worked to model scalpel-liver interaction force through experiments and elastic Finite Element Model analysis in ANSYS but did not examine the effect of varying scalpel geometries. Needle cutting geometry has been well defined in terms of inclination (λ) and rake angles (α) for a variety of styles of needles ^{10, 11}. A study has yet to focus on the work conducted in this paper; defining the geometry of a scalpel cutting edge in terms of λ , α and the contact surface area, A .

The goal of this paper is to define scalpel blade geometry and examine the cutting forces and deflection between commonly used scalpel blades and phantom gel. A generalized analytic model of λ and α is developed and A is measured for four commonly used scalpel blades. Slide cutting experimental results for these four blades in phantom gel are presented and discussed followed by conclusions.

Scalpel Geometry Model

The geometry of a scalpel cutting edge can be characterized by the same parameters used in oblique cutting for machining tools. These parameters include λ and α . In addition to these parameters, tissue cutting force is also dependent on A . Defining the scalpel edge geometry allows for the comparison of different blade numbers based on their unique configuration.

Inclination angle of scalpel blades

The cutting edge shape of industry standard scalpel blades can be expressed as continuously differentiable polynomials, $f(x)$, as shown in Figure 3 and equation (1). Here x is the cut depth of the scalpel blade measured from the scalpel tip, point **O**. The y -direction is perpendicular to the x -direction and is also the cutting direction of the blade

when used in slide cutting. The unit vector in the y -direction is $\mathbf{v}(x) = \{0, 1, 0\}$. The unit vector in the x -direction is $\mathbf{c}(x) = \{1, 0, 0\}$. The vector tangent to $f(x)$ is $\mathbf{s}(x) = \{1, |f'(x)|, 0\}$ (see Figure 3).

$$y = f(x) \tag{1}$$

The inclination angle is a function of x , $\lambda(x)$, and is defined as the angle between the $\mathbf{P_r}$ plane and the $\mathbf{s}(x)$ vector as shown in Figures 3 and 5^{11, 12}. The $\mathbf{P_r}$ plane has the normal vector $\mathbf{v}(x)$. Therefore the inclination angle can be solved as the complimentary angle between $\mathbf{v}(x)$ and the $\mathbf{s}(x)$ as shown in equation (2).

$$\sin(\lambda(x)) = \frac{|\mathbf{s}(x) \cdot \mathbf{v}(x)|}{|\mathbf{s}(x)| |\mathbf{v}(x)|} \tag{2}$$

On solving equation (2) explicitly for $\lambda(x)$ and substituting in the values for $\mathbf{s}(x)$ and $\mathbf{v}(x)$ vectors equation (3) is produced.

$$\lambda(x) = \sin^{-1}\left(\frac{|f'(x)|}{\sqrt{1 + |f'(x)|^2}}\right) \tag{3}$$

This expression can be further simplified as an inverse tangent of the absolute value $f'(x)$ as shown in equation (4).

$$\lambda(x) = \tan^{-1}(|f'(x)|) \quad (4)$$

Rake angle of scalpel blades

The definition of a tool rake angle is the angle between plane $\mathbf{P}_r$ and $\mathbf{A}_\gamma$ measured in plane $\mathbf{P}_n$ when the bottom rake face acts as the reference plane as shown in Figure 4 with vectors $\mathbf{a}'(x)$ and $\mathbf{b}(x)$ ^{11, 12}. However, the x-y plane has been used as the reference in this study for simplifying the analysis. Thus, the rake angle has been computed by finding the angle between vector $\mathbf{a}(x)$ and vector $\mathbf{b}(x)$; where $\mathbf{a}(x)$ is the cross product of $\mathbf{s}(x)$ and $\mathbf{v}(x)$ as shown in equation (5) and $\mathbf{b}(x)$ is the cross product of $\mathbf{s}(x)$ and $\mathbf{n}_\gamma(x)$ as shown in equation (6), where $\mathbf{n}_\gamma(x) = \{-\sin(\frac{\theta}{2})\sin(\tan^{-1}(|f'(x)|)), \sin(\frac{\theta}{2})\cos(\tan^{-1}(|f'(x)|)), \cos(\frac{\theta}{2})\}$ and $\theta(x)$ is the angle ground into the cutting edge; which is the rake angle of the blade overestimated by a value of $\frac{\theta}{2}$ as shown in Figures 4 and 5.

$$\mathbf{a}(x) = \mathbf{s}(x) \times \mathbf{v}(x) = \{0, 0, 1\} \quad (5)$$

$$\begin{aligned}
\mathbf{b}(x) &= \mathbf{s}(x) \times \mathbf{n}_\gamma(x) \\
&= \{ |f'(x)| \cos\left(\frac{\theta}{2}\right), -\cos\left(\frac{\theta}{2}\right), \sin\left(\frac{\theta}{2}\right) \cos(\tan^{-1}(|f'(x)|)) \\
&\quad + |f'(x)| \sin\left(\frac{\theta}{2}\right) \sin(\tan^{-1}(|f'(x)|)) \}
\end{aligned} \tag{6}$$

The rake angle can be calculated as shown in equation (7). This then simplifies to equation (8) where the rake angle is solely dependent on $\theta(x)$.

$$\alpha(x) + \frac{\theta(x)}{2} = \cos^{-1}\left(\frac{\mathbf{a}(x) \cdot \mathbf{b}(x)}{|\mathbf{a}(x)| |\mathbf{b}(x)|}\right) \tag{7}$$

$$\alpha(x) = 90 - \theta(x) \tag{8}$$

Contact surface area

Scalpel blades with different geometries have different surface areas in contact with tissue during cutting for a given cut depth (x). The total tissue cutting force (F_t) can be decomposed into tissue stiffness force (F_s) and tissue friction force (F_f) as shown in Figure 6. The surface area in contact with the tissue is a major contributing factor to F_f . There are numerous analysis techniques available to determine the surface area of the scalpel in

contact with tissue during the cutting process. However, in this study, accurate three dimensional Computer Aided Design (CAD) Models were developed for the scalpel blades being studied as shown in Figure 7. These models were then used to evaluate $A(x)$ based on x (see Figure 6).

Phantom Gel Slide Cutting Experiment

A scalpel cutting experiment was developed to investigate the relationship between cutting forces and the geometry of the #10, #11, #12 and #15 scalpel blades. The experiment consisted of a scalpel blade cutting through phantom gel in a linear, single axis direction at a fixed initial cutting depth to determine the overall cutting force. The setup for the experiment is displayed in Figure 8. The experimental setup comprises of a linear motor (Dunkermotoren, Germany), a six degrees of freedom (DOF) force sensor (Nano17 from ATI Industrial Automation, NC) mounted on a lead screw driven translating base, a scalpel holder, a rigid stand and a clear polycarbonate base that restrains the phantom gel while testing. For all experiments, polyvinyl chloride phantom gel with a 5:1 ratio of plastic to softener (M.F. Manufacturing Company) was used to ensure consistency in the stiffness of the cutting material.

Surgical Carbon Steel scalpel blades made by Havel (Cincinnati, Ohio) (see Figure 1) were mounted on a scalpel holder and the linear motor propelled the blade through the phantom gel material. The x-axis of the scalpels formed a 90 degree angle with the cutting direction (y-axis) in all the tests performed. The initial cutting depth was set to 6 mm. A LabView computer code was written to precisely control the cutting speed at 2.54 mm/s. All the phantom gel samples were constrained at the bottom surface by the means of double sided tape to provide consistent boundary conditions which is crucial for repeatable machining results as illustrated by *Shih et al.*¹³ in machining elastomers.

A total of 12 trials were conducted, three trials for each of the four blades. The cutting force and cutting depth were recorded. The phantom gel samples dimensions were 119 mm x 66 mm x 19 mm. The first cut was made 20.3 mm from the edge of the sample and two successive cuts were made 12.7 mm apart from the first cut allowing for three test cuts per cutting sample. This minimized boundary effects and isolated each cut from the previous cut thus improving the consistency of the tests.

Analysis and Results

Analysis of cutting tool dimensions

The first step in determining the geometry of the blades is to be able to describe the cutting edge of the blade in terms of a continuously differentiable function, $f(x)$ as shown in equation (1). This was accomplished using image processing techniques that trace the exterior boundaries of objects as shown in Figure 9. This pixilated data is converted to length measurements, which can be done by dividing the number of pixels in a known length (n_L) on the blade by the length itself (L). Every pixel coordinate was then divided by the ratio $\frac{n_L}{L}$ to convert the pixel data to length data. The origin, **O**, was placed at the tip of the cutting edge of the scalpel blade as mentioned above. Various polynomial fits were used to generate a function that describes the cutting edge of the blade with a high accuracy, R^2 values were between 0.997 and 0.999. The polynomials that describe the shapes of the four blades studied are shown in Figure 10 and listed in Table 1. This technique can be extended to any blade with a continuously differentiable cutting edge geometry.

The #10 and #15 scalpel blades have the greatest initial positive slope, while the #11 scalpel blade has the smallest slope magnitude. The #11 scalpel blade displays a constant slope on the plot due to its linear cutting edge (see Figure 10). Due to the scythe

shape cutting geometry of the #12 scalpel blade, it is the only blade examined in this experiment with a negative initial slope. Scalpel #15 is the only examined blade that gradually changes its curvature from convex to concave along its cutting edge. This illustrates how blades geometries can greatly vary and therefore will display a variety of slide cutting characteristics.

The value of θ was measured using a Zeiss Axio Optical Microscope and results are shown in Table 1. This value of θ changed minimally in relationship to x ; therefore, it was assumed in the calculations of λ and α that θ was constant.

Table 1. List of scalpel cutting edge polynomials and measured ground angles

Blade #	Polynomial $f(x)$ (x in mm)	Average $\theta \pm \sigma$ (Degrees)
10	$7.810 \cdot 10^{-3} x^3 - 2.071 \cdot 10^{-1} x^2 + 1.831 x$	26.43 ± 0.74
11	$2.658 \cdot 10^{-1} x$	14.07 ± 2.40
12	$3.965 \cdot 10^{-2} x^2 - 8.523 \cdot 10^{-1} x$	20.39 ± 1.36
15	$-1.313 \cdot 10^{-3} x^4 + 4.119 \cdot 10^{-2} x^3 - 4.509 \cdot 10^{-1} x^2 + 1.964 x$	27.53 ± 1.07

Analysis of cutting depth

The initial cut depth for all the slide cutting experiments was 6 mm; however, the compliance of the phantom gel caused this value to change continuously as the scalpel cut through tissue. Also, different scalpel geometries led to different amounts of tissue deflection which affects the value of F_t . Thus, to closely estimate the value of the cut depth at each point in the scalpel's trajectory through the tissue, ink was poured into the top of the cut groove after each cut was complete. Once the ink seeped all the way through the cut and dried, a knife was used to separate a section of the tissue containing the cross section of the scalpel cut which showed up in the dried black ink as shown in Figure 11.

After this, the top and bottom surfaces of the cut were digitized similarly to the previous method used to measure the scalpel cutting surface. After digitization, the pixel data was converted to length data and then the vertical difference between the two curves is the cut depth or x value as a function of cut length (y).

Results of geometric model

The value of θ has no effect on the value of $\lambda(x)$ since the two parameters are unrelated, however, it does have a major influence on $\alpha(x)$. A lower θ and therefore a

sharper blade leads to an increase in the angle that its rake face vector $\mathbf{b}(x)$ makes with the vector vertical to the x - y plane $\mathbf{a}(x)$ to give higher rake angle values.

Equations (4) and (8) are used to calculate $\lambda(x)$ and $\alpha(x)$ (see Figures 12 and 13). The accuracy of these models were verified using the CAD models developed for the four scalpels. The results of the $A(x)$ were measured using the CAD models of the scalpel blades (see Figure 14).

Figure 12 displays the inclination angles of all four scalpel blades as a function of cutting depth x . The #10 and #15 scalpel blades have high initial inclination angles at point $\mathbf{O}$ (x is zero). However, they also exhibit a steep drop in the inclination angle which quickly reduces to values lower than scalpels #12 and #11 for higher cutting depths. Scalpel #15 reaches an inclination angle of zero at $x = 4$ mm due to the change in curvature from convex to concave. The #11 scalpel blade, shows a zero slope because of its linear cutting edge geometry giving it a constant inclination angle. For scalpel #12, the inclination angle also decreased, but more gradually compared to #10 and #15.

Figure 13 displays the rake angle of each of the four scalpel blades used in this study. A high rake angle signifies a sharper scalpel cutting edge surface. Irrespective of the shape of the cutting edge, the rake angle for scalpel blades remains constant as it is by definition measured in the plane $\mathbf{P}_n$ which always stays normal to the $\mathbf{s}(x)$ vector (see

Figure 5). Equation (8) captures this behavior as α is only a function of the ground angle or θ which has been assumed to be constant in this study and hence α also displays a constant value. Blade #11 displays the highest value of rake angle followed by blade #12, blade #10 and finally blade #15 which has the smallest rake angle of the four blades studied.

Figure 14 displays the contact surface area of each for the scalpel blades used in the experiment during slide cutting through the phantom gel sample as a function of the cut depth (x). All plots have a similar upward trend as a result of increasing contact surface area as cutting depth increases. At the full 10 mm of insertion scalpel #10 has the highest contact area followed by #12, #15 and #11. At $x < 4.66$ mm of insertion scalpel #12 has the highest amount of contact area. These differing scalpel geometries result in different contact areas between scalpel and tissue. Scalpels with higher surface area will increase the friction between the scalpel and the tissue and therefore lead to higher F_t values.

Results of slide cutting experiments

Figure 15 depicts the results of the total cutting force averages obtained over the three cuts made per sample in the test method described previously. The varying geometry of the four blades creates unique cutting force profiles for each of the blades. All four graphs have an initial spike as expected due to the fracture mechanism required at the

beginning of the cut. However, they all settle to a relatively steady value once x becomes fairly constant by $y > 30$ mm. Though the initial cut depth was 6 mm for all samples, the average cut depth in the steady force region varied for the four blades based on how their geometry interacted with the phantom gel and the amount of deflection it caused during slide cutting.

Table 2 depicts the average values of x , $A(x)$, $\lambda(x)$ and $\alpha(x)$ along cut length values (y) with a relatively steady cutting force region which is taken to be where y is between 40 mm and 100 mm (see Figure 15). Blade #10 displays the lowest average F_t value followed by blade #15, #11 and #12 respectively. Using the data from the scalpel cut digitization along with the F_t graph, one can normalize the F_t with respect to x as shown in Figure 16. Blade #10 displays the lowest average normalized F_t value followed by blade #15, #12 and #11 respectively.

In conventional cutting a lower cutting force is able to be obtained by reducing contact area and increasing rake and inclination angle. This trend appears to also be true in scalpel tissue cutting as shown in Table 2. It is shown that Blade #10 contains the highest average inclination angle and the lowest average cutting force. Blade #12 contains the highest contact area and the highest average cutting force.

Table 2. Average values of crucial cutting parameters in the steady cutting force region

Blade #	Average x (mm)	Average $A(x)$ (mm²)	Average $\lambda(x)$ (Deg.)	Average $\alpha(x)$ (Deg.)	Average F_t (N)
10	4.29	29.37	45.33	63.57	0.52
11	5.62	19.51	14.88	75.93	0.79
12	6.48	52.14	30.32	69.61	1.17
15	4.36	23.57	28.78	62.47	0.57

Conclusions

In this study, a generalized geometric model that describes the cutting edge of scalpel blades was developed. This study specifically focused on four industry standard scalpel blades #10, #11, #12 and #15. Analytical solutions to determine the inclination angle (λ) and rake angle (α) have been found and are expressed in equations (4) and (8) respectively. CAD models of the four blades were developed to estimate the contact surface area (A) between the scalpel and the phantom gel during slide cutting experiments on phantom gel.

Through the results of the analytical expressions developed, it is evident that the geometries of the four blades studied have various similarities and differences. These characteristics lead to the differences in the interaction between the scalpel and phantom gel during the slide cutting experiments. Further investigation is required in order to model the exact relation between these parameters and the cutting force. The generalized analytical solutions developed to calculate the crucial cutting parameters provides for the first time, a quantitative technique to compare any scalpel blade cutting geometries.

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Appendix

Notation

$\mathbf{a}$	Vector formed by intersection of planes $\mathbf{P_n}$ and $\mathbf{P_r}$
A	Contact surface area between scalpel and tissue
$\mathbf{A}_\gamma$	Plane with normal vector $\mathbf{n}_\gamma$
$\mathbf{b}$	Vector formed by intersection of planes $\mathbf{P_n}$ and $\mathbf{A}_\gamma$
$\mathbf{c}$	Unit vector along cut depth x
f	Continuously differentiable function describing the cutting edge geometry of the scalpel
F_t	Total scalpel slide cutting force
F_s	Stiffness scalpel slide cutting force
F_f	Friction scalpel slide cutting force
L	Arbitrary known length on scalpel geometry
n_L	Number of pixels in a known length $\mathbf{L}$
$\mathbf{n}_\gamma$	Normal vector to scalpel rake face

O	Origin of x - y plane and scalpel cutting edge tip
P_n	Plane with normal vector s
P_r	Plane with normal vector v
s	Vector tangent to f
v	Unit vector in cutting direction y
x	Scalpel cut depth
y	Scalpel cut length and slide cutting direction
α	Rake angle of scalpel cutting edge
λ	Inclination angle of scalpel cutting edge
θ	Angle ground by scalpel manufacturer onto cutting edge

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Figure 1. Commonly used scalpel blades

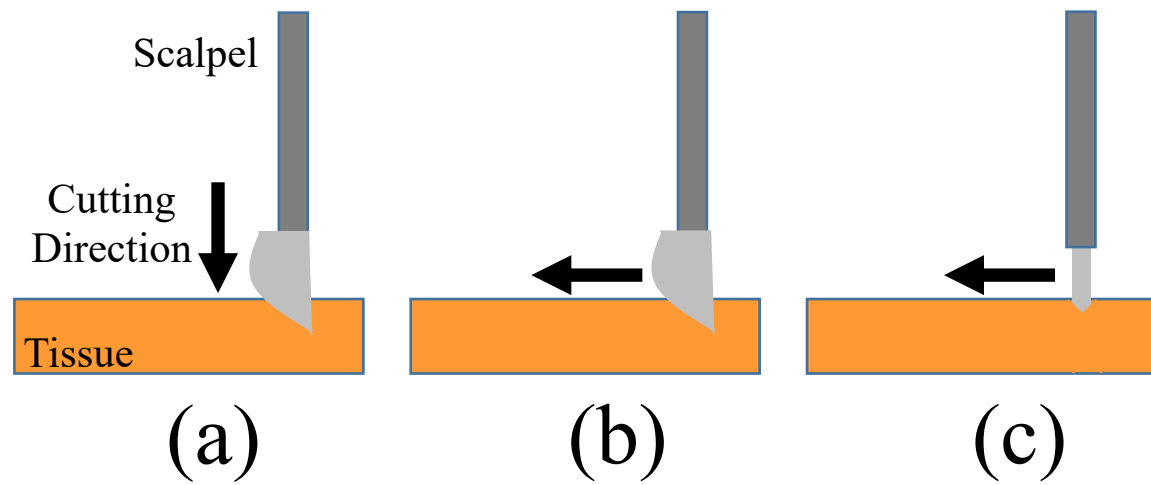


Figure 2. Tissue cutting techniques of (a) press cutting, (b) slide cutting and (c) scrape cutting

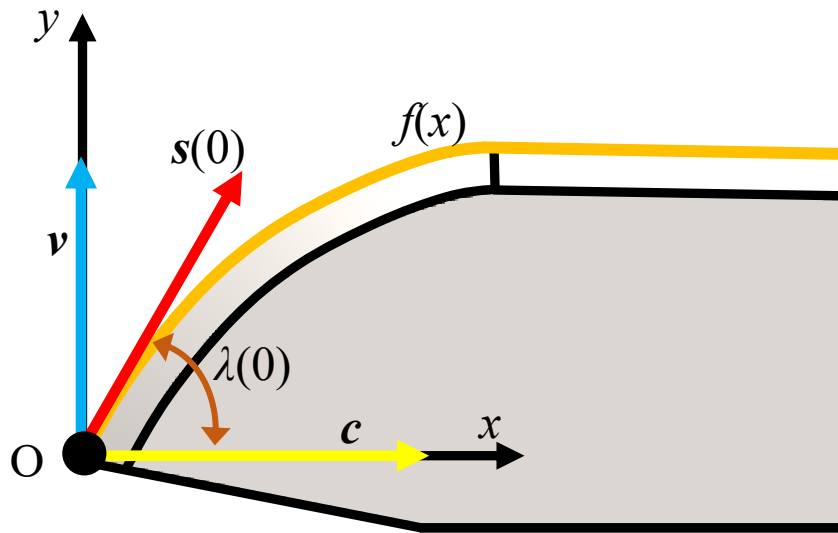


Figure 3. Coordinate system and vectors used in the geometric model to define the inclination angle of scalpels

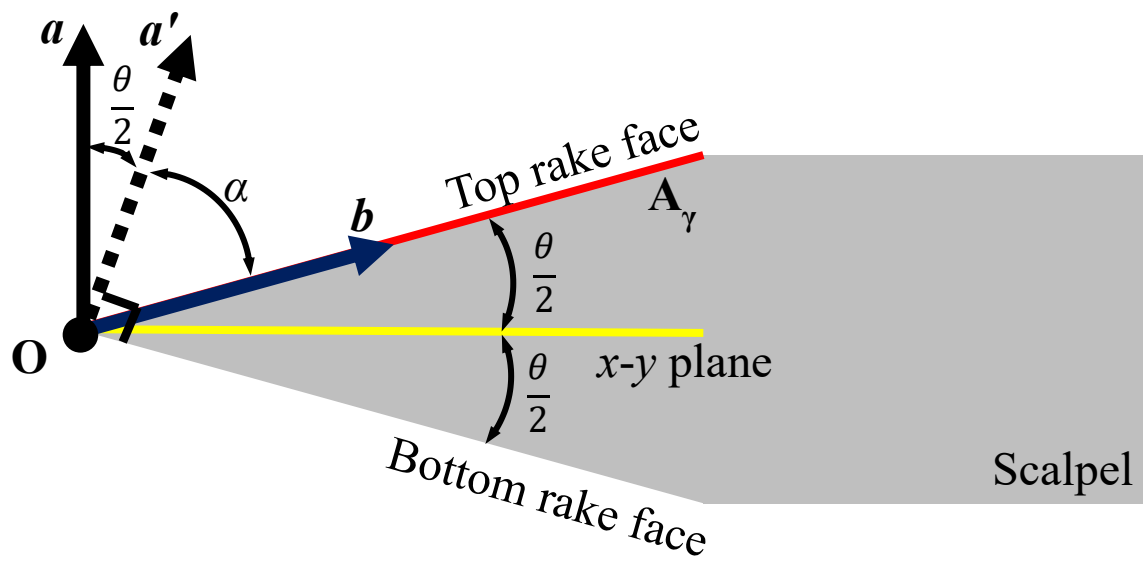


Figure 4. Cross section of scalpel blade displaying the vectors used in the geometric model of the rake angle

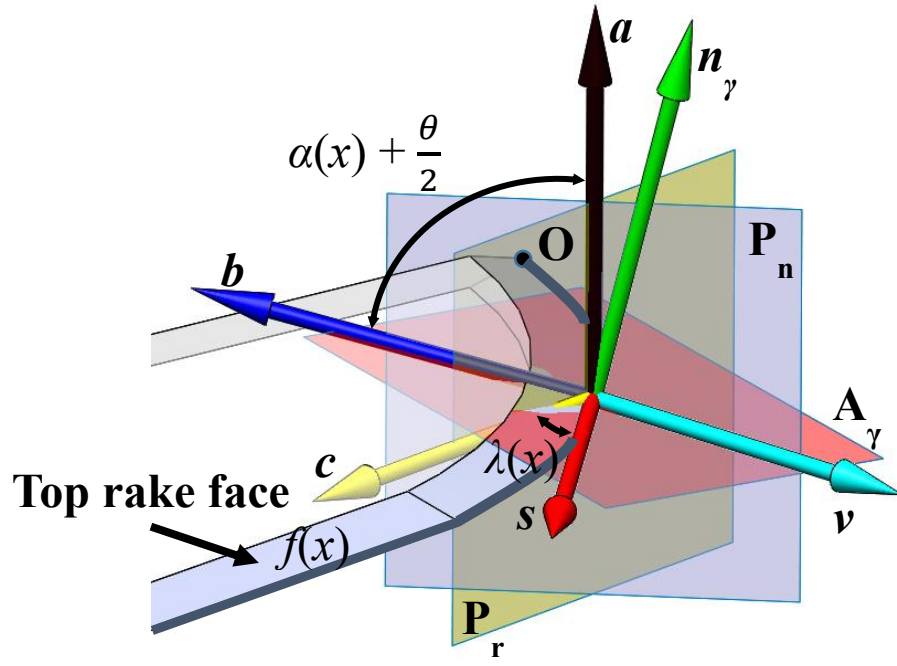


Figure 5. Coordinate system and vectors used in the geometric model to define the rake angle of scalpels

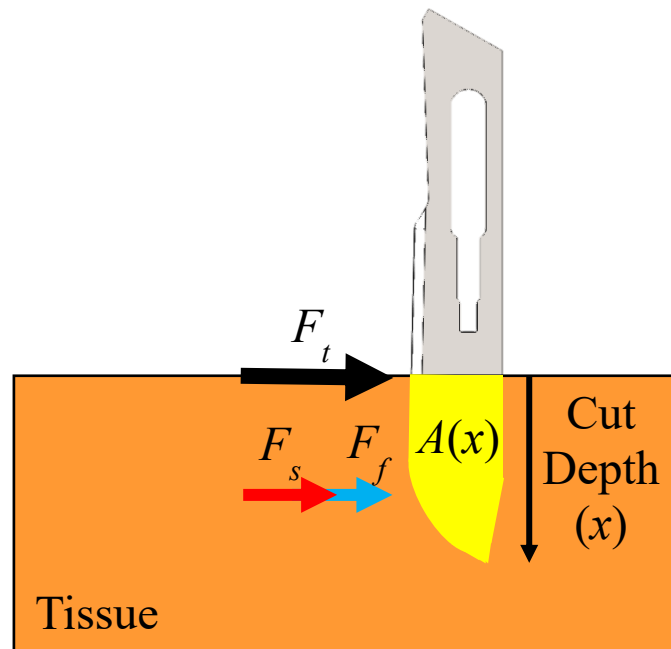


Figure 6. Force decomposition and contact area between scalpel and tissue during slide cutting

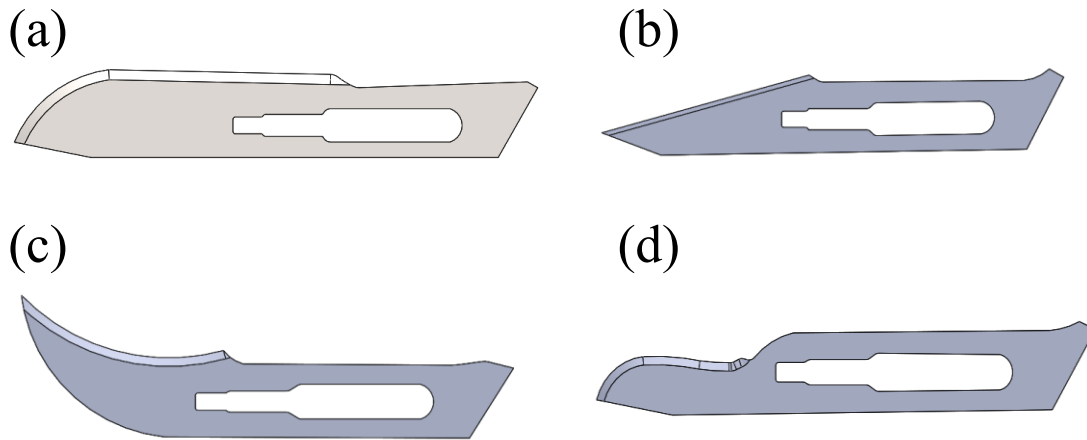


Figure 7. CAD model of scalpel blades (a) #10 (b) #11 (c) #12 (d) #15

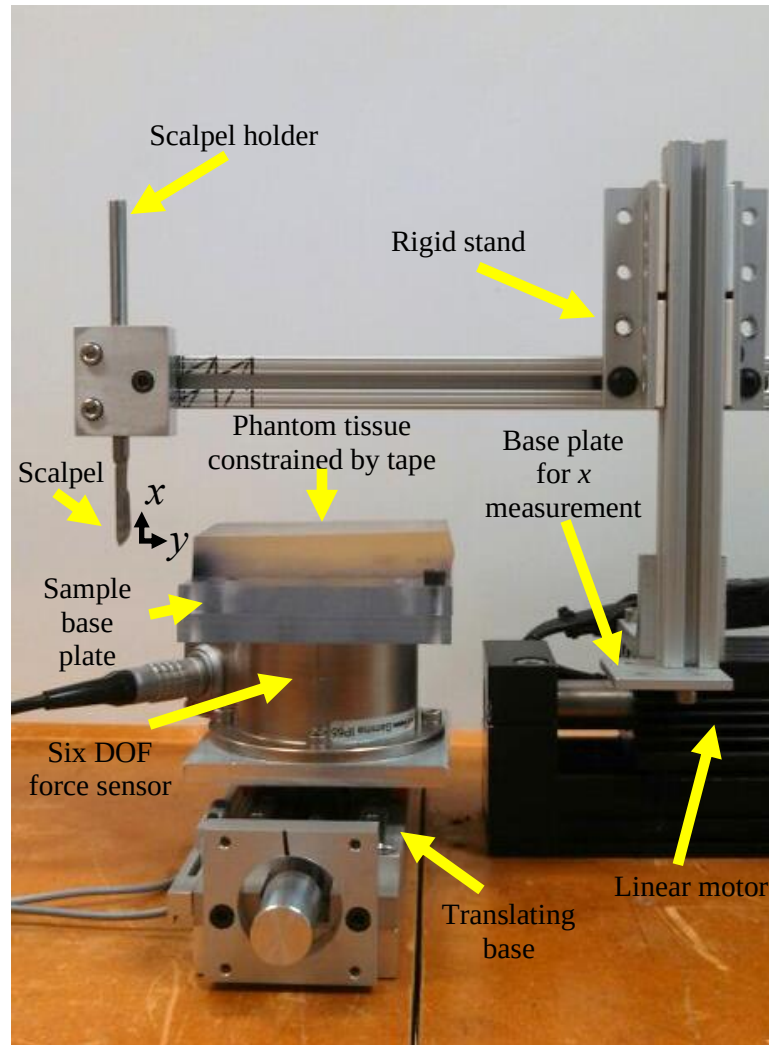


Figure 8. Experimental setup for slide cutting of phantom gel

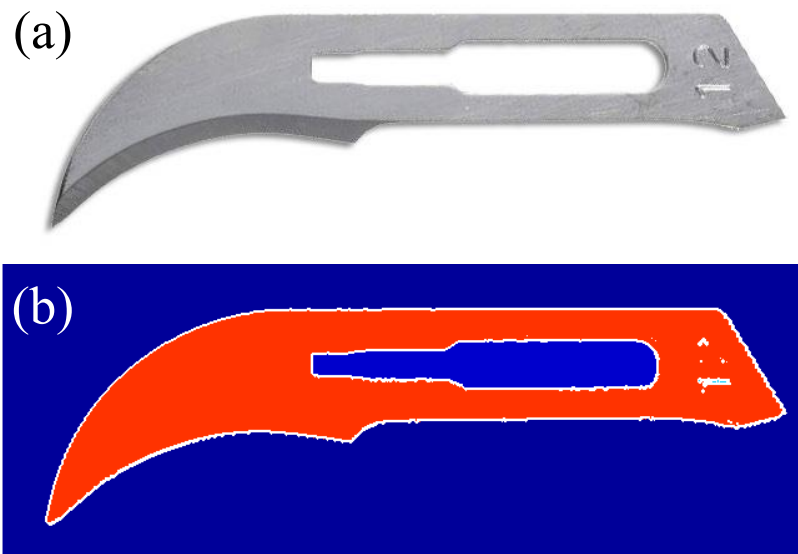


Figure 9. (a) Digital image of scalpel #12 (b) Digitized boundary of scalpel #12 depicted by white pixels

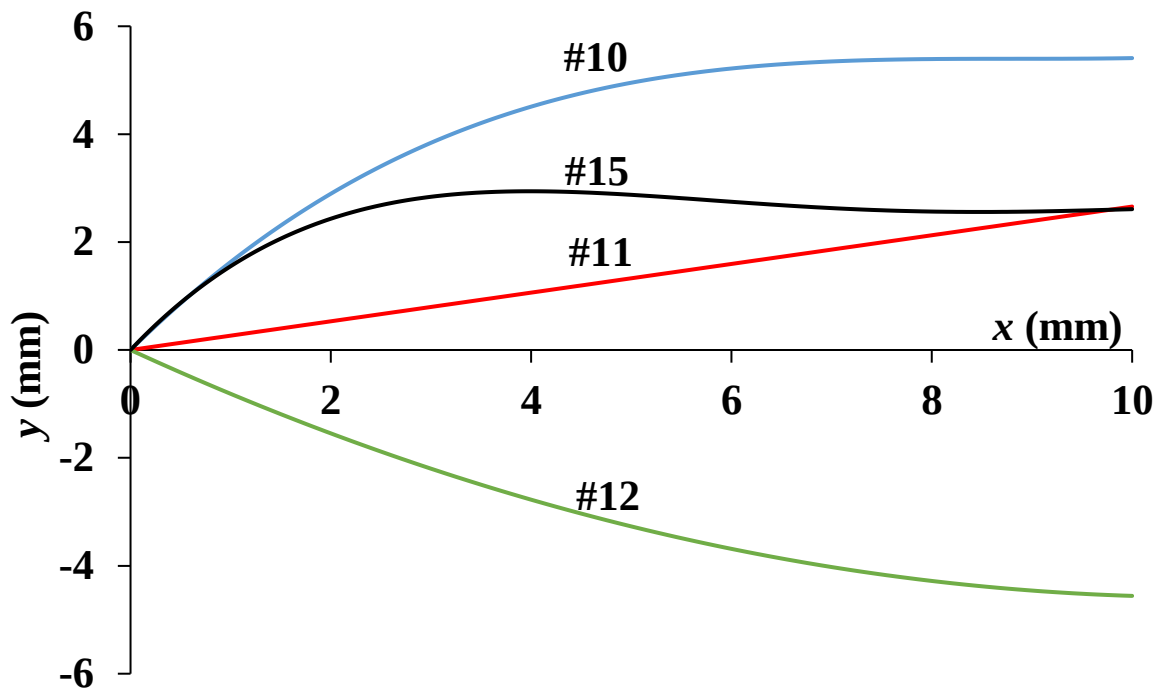


Figure 10. Cutting edge geometry of scalpels used in this study

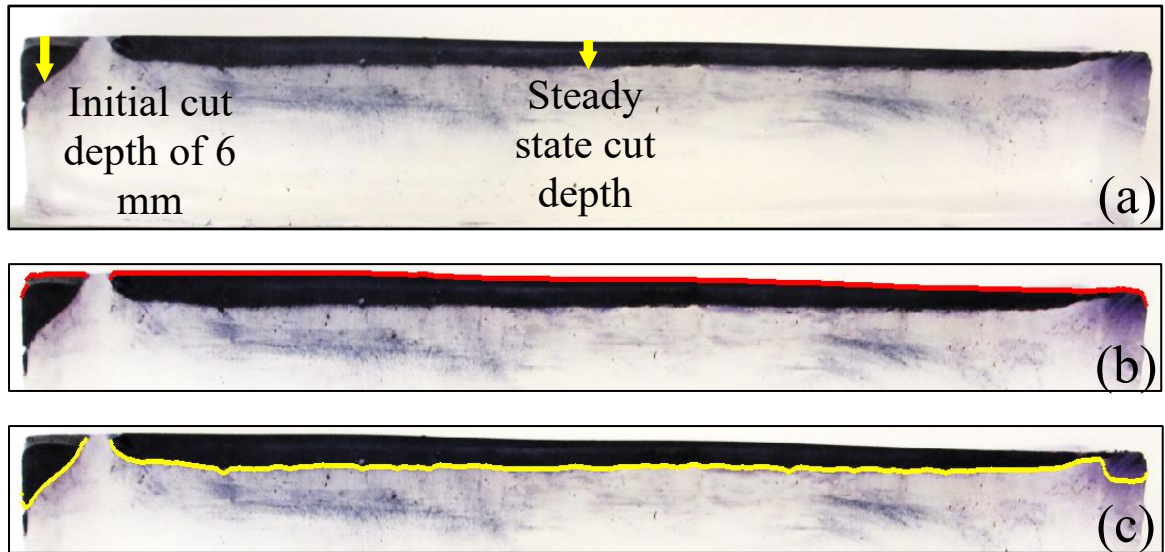


Figure 11. Section of phantom gel showing (a) cut pattern used to estimate cut depth (b) digitization of top surface of cut pattern (c) digitization of bottom surface of cut pattern

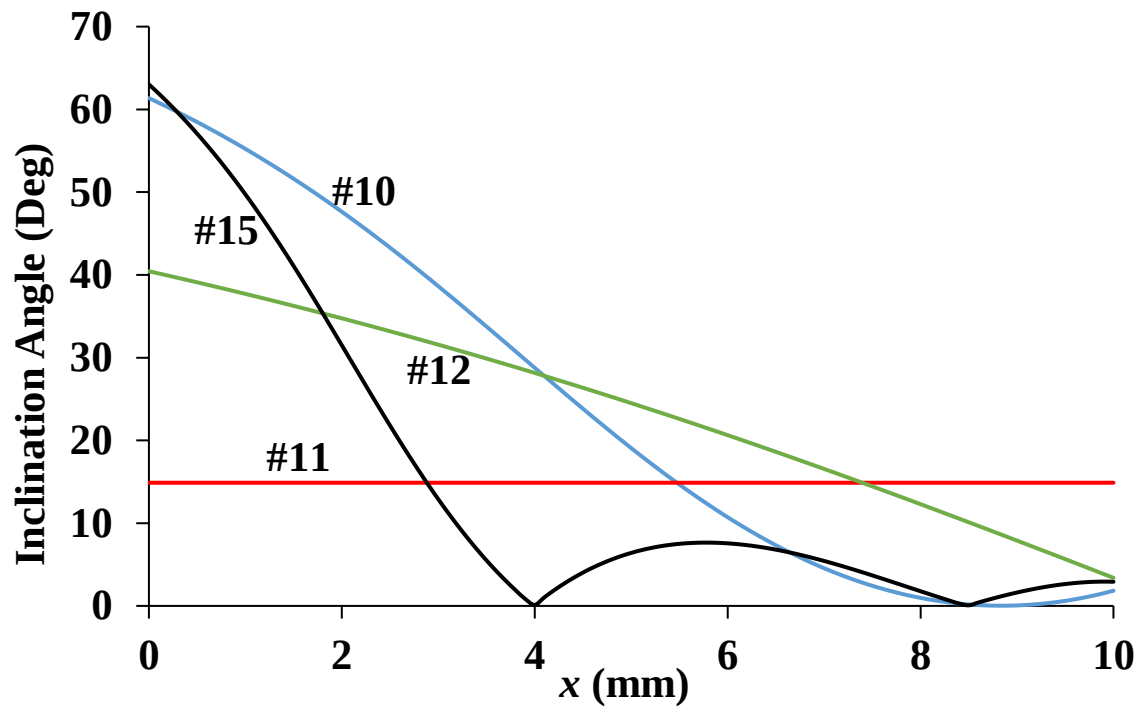


Figure 12. Inclination angle estimation as a function of cutting depth

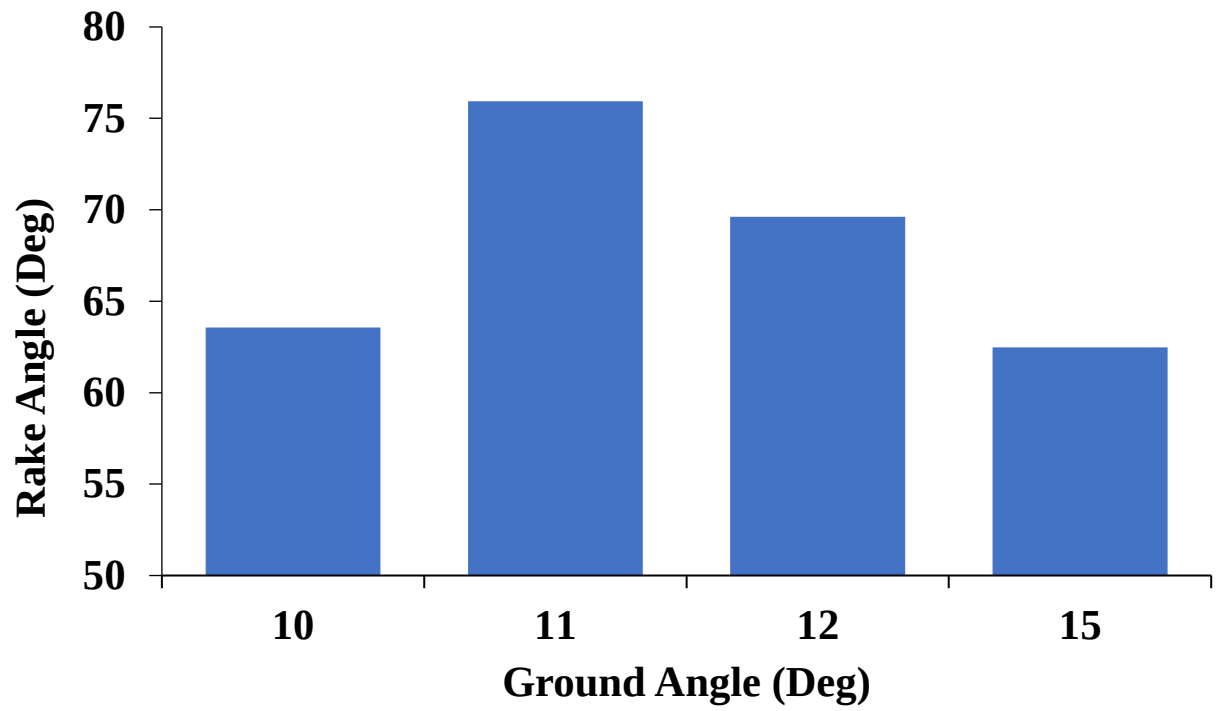


Figure 13. Bar graph of rake angle estimation

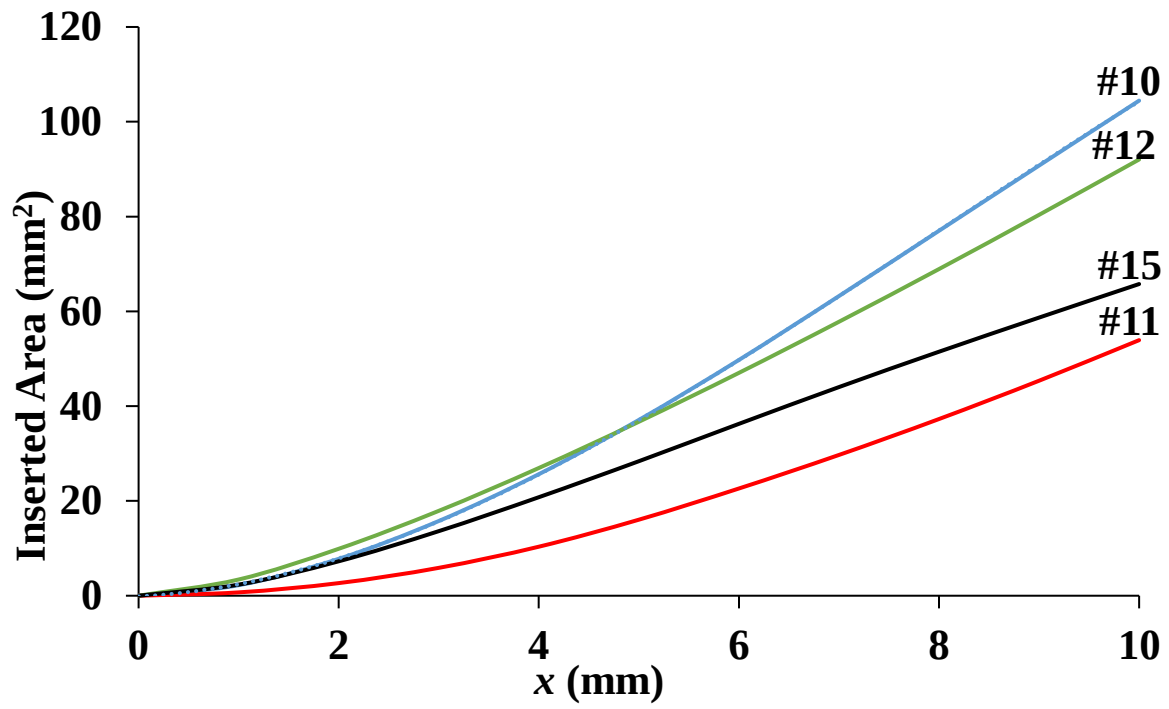


Figure 14. Contact surface area between tissue and scalpel during slide cutting as a function of cutting depth

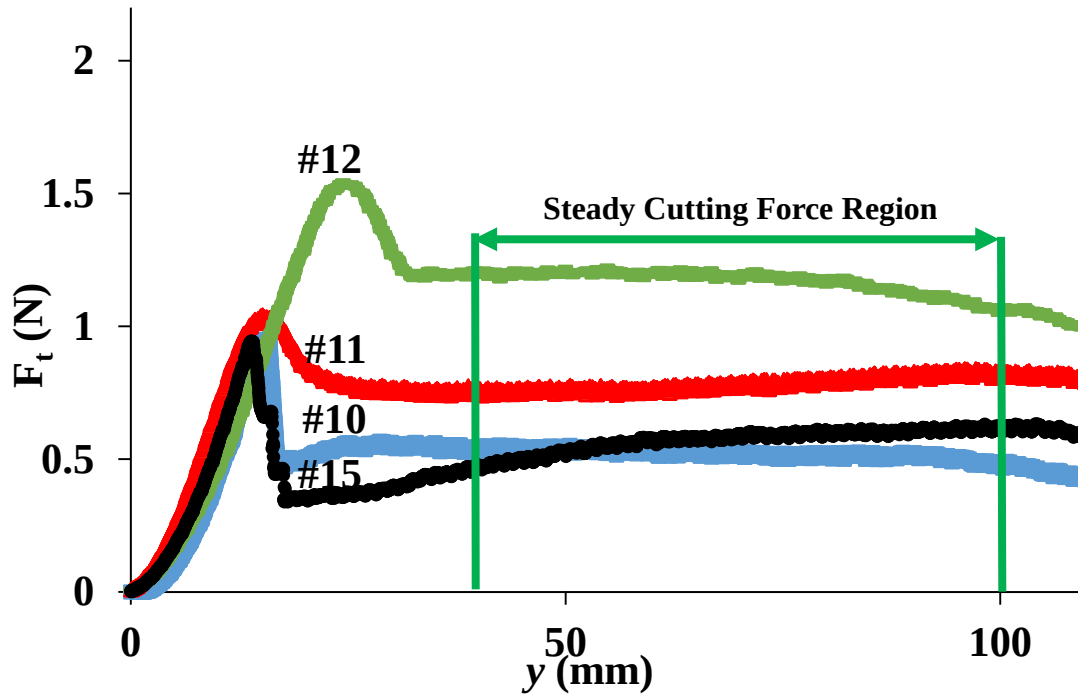


Figure 15. Total cutting force during slide cutting as a function of cut length

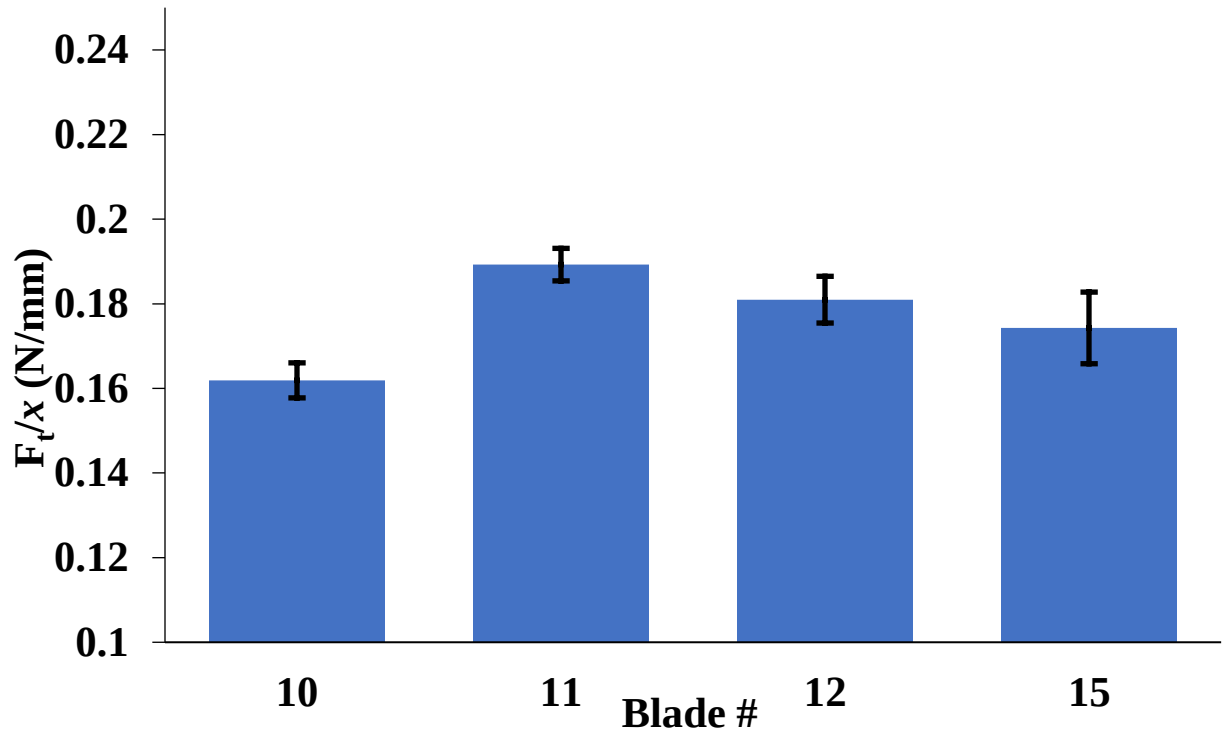


Figure 16. Bar plot of normalized total cutting force with its standard deviation